

Personal Information

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Github: <https://github.com/Brebber/Thesis>

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Data Context

The dataset consists of two datasets from rijdendetreinen.nl, the services dataset from 2019-2024 and the disruption datasets from these same years. These datasets have been appended and transformed so we have a daily overview over whether a trajectory has been disrupted that day. Additionally, the daily weather data has been sourced from KNMI, for a few different weather stations. This has been merged with the daily trajectory dataset based on the closest weather station to the source of the trajectory. For the train stations coordinates, the stations-2019-nl dataset has been used, also provided by rijdendetreinen.nl.

The main dataset has already been preprocessed, all the trajectories have been grouped by day, the daily weather data has been added. Apart from the data, origin and destination of each trajectory, the dataset contains the number of rides planned, the total final arrival delay of that day, the amount of times a final arrival has been cancelled, the amount of times the trajectory has been completely cancelled and the total amount of intermediate arrival delays. From the disruptions dataset, a list of the statistical causes of the disruptions that occurred that day has been added, as well as the total amount of disruptions that day and a binary variable that is true if the trajectory has been disrupted at least once that day. The added weather variables are:

FHVEC : Vector mean windspeed (in 0.1 m/s)

TG : Daily mean temperature in (0.1 degrees Celsius)

TN : Minimum temperature (in 0.1 degrees Celsius)

TX : Maximum temperature (in 0.1 degrees Celsius)

T10N : Minimum temperature at 10 cm above surface (in 0.1 degrees Celsius)

SQ : Sunshine duration (in 0.1 hour) calculated from global radiation (-1 for <0.05 hour)

DR : Precipitation duration (in 0.1 hour)

RH : Daily precipitation amount (in 0.1 mm) (-1 for <0.05 mm)

RHX : Maximum hourly precipitation amount (in 0.1 mm) (-1 for <0.05 mm)

VVX : Maximum visibility

Data Description

```
In [45]: # Imports
import os
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import classification_report, confusion_matrix
```

Data Loading

```
In [ ]: # Load main dataset
disruptions_df = pd.read_csv('daily_disruptions_weather.csv')
disruptions_df['Date'] = pd.to_datetime(disruptions_df['Date'])

# Load data necessary for graph
coordinates_df = pd.read_csv('stations-2023-09-nl.csv')
distances_df = pd.read_csv('station_distances.csv')

# Merge distance data
disruptions_df = disruptions_df.merge(
    distances_df,
    how='left',
    left_on=['source', 'target'],
    right_on=['origin_name', 'destination_name']
)
disruptions_df['distance'] = disruptions_df['distance'].fillna(0)
disruptions_df = disruptions_df.drop(columns=['Unnamed: 0', 'origin_name', 'destination_name'])
```

disruptions_df

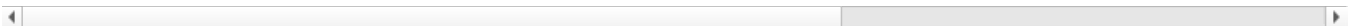
C:\Users\brake\AppData\Local\Temp\ipykernel_7304\1813439240.py:2: DtypeWarning:

Columns (10,11,12,13,14,15,16,18,19) have mixed types. Specify dtype option on import or set low_memory=False.

Out[]:

	Date	source	target	Rides planned	Final arrival delay	Final arrival cancelled	Completely cancelled	Intermediate arrival delays	Statistical Causes	Disruption Count	...	SQ	T
0	2019-01-01	's-Hertogenbosch	Arnhem Centraal	1	0	0	0	1	No Disruption	0.0	...	7	7
1	2019-01-01	's-Hertogenbosch	Den Haag Centraal	33	4	2	0	27	No Disruption	0.0	...	7	7
2	2019-01-01	's-Hertogenbosch	Deurne	17	4	1	0	11	No Disruption	0.0	...	7	7
3	2019-01-01	's-Hertogenbosch	Dordrecht	11	0	0	0	3	No Disruption	0.0	...	7	7
4	2019-01-01	's-Hertogenbosch	Eindhoven Centraal	18	2	0	0	7	No Disruption	0.0	...	7	7
...	...	...	...	...	...	...	...	...	...	...	...	...	...
654431	2024-12-31	Zwolle	Nijmegen	1	0	0	0	1	No Disruption	0.0	...	0.0	52
654432	2024-12-31	Zwolle	Roosendaal	31	5	6	0	28	No Disruption	0.0	...	0.0	52
654433	2024-12-31	Zwolle	Schiphol Airport	1	0	0	0	0	No Disruption	0.0	...	0.0	52
654434	2024-12-31	Zwolle	Utrecht Centraal	29	1	0	0	19	No Disruption	0.0	...	0.0	52
654435	2024-12-31	Zwolle	Zutphen	1	0	0	0	1	No Disruption	0.0	...	0.0	52

654436 rows × 22 columns



Data Imbalance

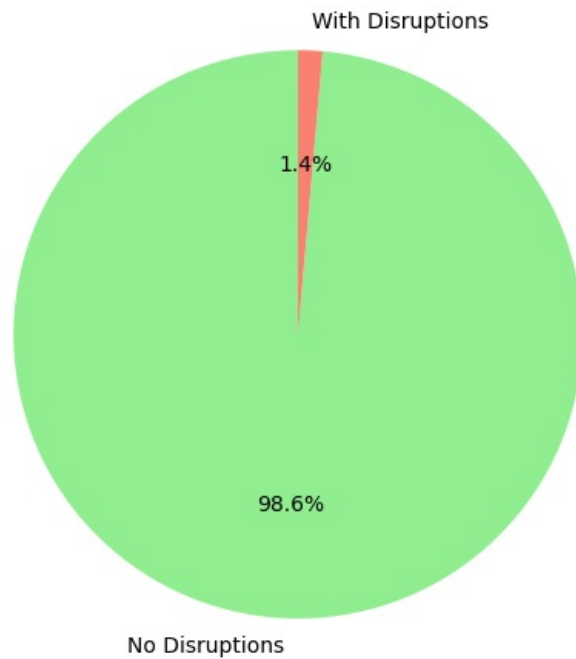
```
In [19]: import matplotlib.pyplot as plt

# Count the number of trajectories with Disruption Count == 0 and > 0
disrupted = (disruptions_df["Disrupted"] == True).sum()
non_disrupted = (disruptions_df["Disrupted"] == False).sum()

# Data for the pie chart
labels = ['No Disruptions', 'With Disruptions']
sizes = [non_disrupted, disrupted]
colors = ['lightgreen', 'salmon']

# Plot the pie chart
plt.figure(figsize=(8, 6))
plt.pie(sizes, labels=labels, autopct='%1.1f%%', startangle=90, colors=colors)
plt.title('Trajectories with and without Disruptions')
plt.show()
```

Trajectories with and without Disruptions



Most Common Statistical Causes Over Time

```
In [20]: disrupted_df = disruptions_df[disruptions_df['Disrupted'] == True]
disrupted_df['Year'] = pd.to_datetime(disrupted_df['Date']).dt.year

# Explode the 'Statistical Causes' column to separate individual causes
disrupted_df_exploded = disrupted_df.copy()
disrupted_df_exploded['Statistical Causes'] = disrupted_df_exploded['Statistical Causes'].apply(eval) # Convert
disrupted_df_exploded = disrupted_df_exploded.explode('Statistical Causes')

# Group by year and cause, then count occurrences
causes_by_year = disrupted_df_exploded.groupby(['Year', 'Statistical Causes']).size().unstack(fill_value=0)

# Create subplots for all years in the same figure
fig, axes = plt.subplots(nrows=2, ncols=3, figsize=(20, 12), sharey=True)

# Flatten the axes array for easier iteration
axes = axes.flatten()

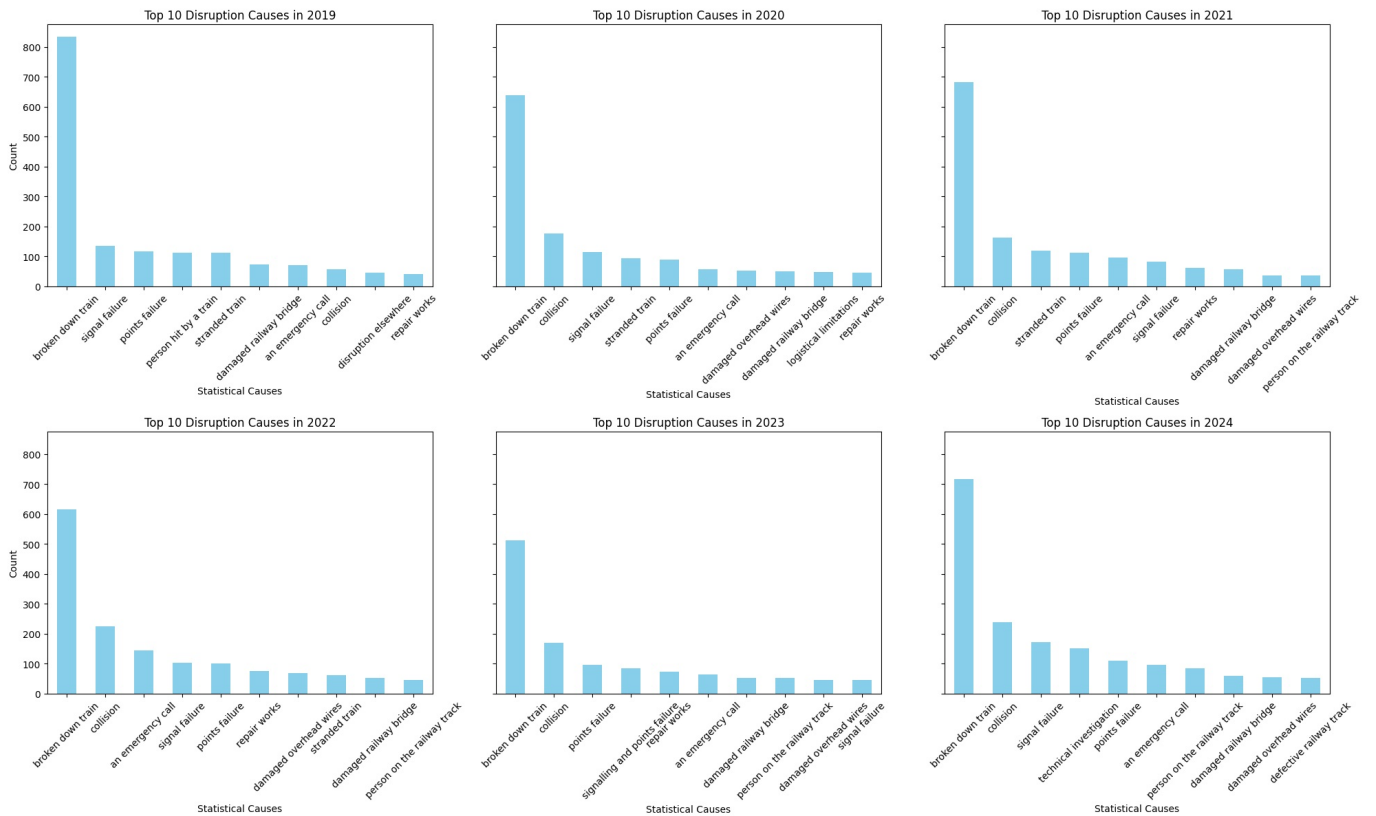
# Plot each year's data in a separate subplot
for i, year in enumerate(causes_by_year.index):
    ax = axes[i]
    causes_by_year.loc[year].nlargest(10).sort_values(ascending=False).plot(kind='bar', color='skyblue', ax=ax)
    ax.set_title(f'Top 10 Disruption Causes in {year}')
    ax.set_xlabel('Statistical Causes')
    ax.set_ylabel('Count')
    ax.tick_params(axis='x', rotation=45)

# Adjust layout to prevent overlap
plt.tight_layout()
plt.show()
```

C:\Users\brake\AppData\Local\Temp\ipykernel_7304\2703092600.py:2: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.
Try using `.loc[row_indexer,col_indexer] = value` instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy



Outliers

```
In [21]: top_10_disruptions = disruptions_df.sort_values(by='Disruption Count', ascending=False)[['Date', 'source', 'target', 'Disruption Count']]
top_10_disruptions
```

```
Out[21]:
```

	Date	source	target	Disruption Count
97240	2019-12-02	Amsterdam Centraal	Schiphol Airport	9.0
489037	2023-07-05	Alkmaar	Amsterdam Centraal	7.0
117997	2020-02-08	Amsterdam Centraal	Utrecht Centraal	7.0
376979	2022-06-27	Amsterdam Centraal	Schiphol Airport	6.0
370671	2022-06-07	's-Hertogenbosch	Utrecht Centraal	6.0
96362	2019-11-29	Amsterdam Centraal	Utrecht Centraal	6.0
1932	2019-01-08	Almelo	Deventer	5.0
121446	2020-02-19	Amsterdam Centraal	Schiphol Airport	5.0
489737	2023-07-07	Amsterdam Centraal	Haarlem	5.0
80299	2019-10-08	Amsterdam Centraal	Schiphol Airport	5.0

```
In [22]: summed_disruptions = disruptions_df.groupby(['source', 'target'])[['Rides planned', 'Disruption Count']].sum()
summed_disruptions = summed_disruptions.sort_values(by='Disruption Count', ascending=False)

# Display the top 10 most disrupted trajectories
summed_disruptions.head(10)
```

Out[22]:

	source		target	Rides planned	Disruption Count
680	Amsterdam Centraal		Utrecht Centraal	10012	847.0
59	's-Hertogenbosch		Utrecht Centraal	34155	784.0
1656	Den Haag Centraal		Utrecht Centraal	36528	758.0
131	Alkmaar	Amsterdam Centraal		16234	705.0
1337	Breda		Dordrecht	7402	653.0
2523	Groningen		Zwolle	67157	542.0
1025	Arnhem Centraal		Nijmegen	20816	495.0
613	Amsterdam Centraal		Haarlem	8850	481.0
3908	Roosendaal		Vlissingen	18700	454.0
1979	Dordrecht	Rotterdam Centraal		10610	445.0

In [23]:

```
# Display the 10 least disrupted trajectories
summed_disruptions = summed_disruptions.sort_values(by='Disruption Count', ascending=True)
summed_disruptions.head(10)
```

Out[23]:

	source		target	Rides planned	Disruption Count
4403	Tilburg	Den Haag Centraal		827	0.0
18	's-Hertogenbosch		Deventer	117	0.0
39	's-Hertogenbosch		Leiden Centraal	11865	0.0
84	't Harde		Ermelo	2	0.0
40	's-Hertogenbosch		Maastricht	984	0.0
43	's-Hertogenbosch		Nijmegen Goffert	3	0.0
67	's-Hertogenbosch		Vught	32	0.0
68	's-Hertogenbosch		Weert	5	0.0
17	's-Hertogenbosch		Deurne	55874	0.0
69	's-Hertogenbosch		Wijchen	1354	0.0

Pearson's Correlation Between Variables

In [24]:

```
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
import seaborn as sns

# Strip column names to remove leading/trailing spaces
disruptions_df.columns = disruptions_df.columns.str.strip()

# Select weather-related features
weather_features = ['RH', 'SQ', 'TG', 'TN', 'TX', 'RHX', 'VVX', 'T10N', 'FHVEC']
weather_data = disruptions_df[weather_features].dropna()

# Clean the data: strip spaces and coerce non-numeric values to NaN
weather_data = weather_data.applymap(lambda x: str(x).strip() if isinstance(x, str) else x)
weather_data = weather_data.apply(pd.to_numeric, errors='coerce').dropna()
# Standardize the data
scaler = StandardScaler()
weather_data_scaled = scaler.fit_transform(weather_data)

# Add the 'Disrupted' column and other specified columns to the weather data for correlation analysis
columns_to_add = ['Rides planned', 'Final arrival delay', 'Final arrival cancelled', 'Completely cancelled', 'I
for column in columns_to_add:
    weather_data[column] = disruptions_df.loc[weather_data.index, column].astype(float)

# Compute Pearson's correlation coefficients
correlation_matrix = weather_data.corr(method='pearson')[['Disruption Count']]
print(correlation_matrix)
```

C:\Users\brake\AppData\Local\Temp\ipykernel_7304\2382102600.py:13: FutureWarning:

DataFrame.applymap has been deprecated. Use DataFrame.map instead.

	Disruption Count
RH	0.001742
SQ	0.003050
TG	0.006394
TN	0.005256
TX	0.006743
RHX	0.000621
VVX	0.001177
T10N	0.003777
FHVEC	0.005219
Rides planned	-0.015293
Final arrival delay	-0.010777
Final arrival cancelled	0.000560
Completely cancelled	0.002781
Intermediate arrival delays	-0.022565
distance	-0.065018
Disruption Count	1.000000

Graph of the overall Network with all Possible Edges

```
In [38]: import plotly.graph_objects as go
import networkx as nx

sources = disruptions_df['source'].unique()
coordinates_df = pd.read_csv('stations-2023-09-nl.csv')
destinations = disruptions_df['target'].unique()
stations = list(set(sources).union(set(destinations)))
edges = disruptions_df[['source', 'target']].values

G = nx.Graph()
G.add_nodes_from(stations)
G.add_edges_from(edges)

# Create a Plotly figure
# Filter coordinates_df to include only stations present in the graph
coordinates_df = coordinates_df[coordinates_df['name_long'].isin(G.nodes())]

# Create a dictionary of positions based on coordinates
pos = {row['name_long']: (row['geo_lng'], row['geo_lat']) for _, row in coordinates_df.iterrows()}

# Remove nodes from the graph that are not in the coordinates_df
nodes_to_remove = [node for node in G.nodes() if node not in pos]
G.remove_nodes_from(nodes_to_remove)

# Extract edge positions
edge_x = []
edge_y = []
for edge in G.edges():
    x0, y0 = pos[edge[0]]
    x1, y1 = pos[edge[1]]
    edge_x.extend([x0, x1, None])
    edge_y.extend([y0, y1, None])

# Create edge trace
edge_trace = go.Scatter(
    x=edge_x, y=edge_y,
    line=dict(width=0.5, color='#888'),
    hoverinfo='none',
    mode='lines'
)

# Extract node positions and attributes
node_x = []
node_y = []
for node in G.nodes():
    x, y = pos[node]
    node_x.append(x)
    node_y.append(y)

# Create node trace
node_trace = go.Scatter(
    x=node_x, y=node_y,
    mode='markers',
    hoverinfo='text',
    marker=dict(
        showscale=True,
        colorscale='YlGnBu',
        size=10,
        colorbar=dict(
            thickness=15,
            title=dict(
                text='Node Connections',
```

```

        side='right'
    ),
    xanchor='left'
)
)
)

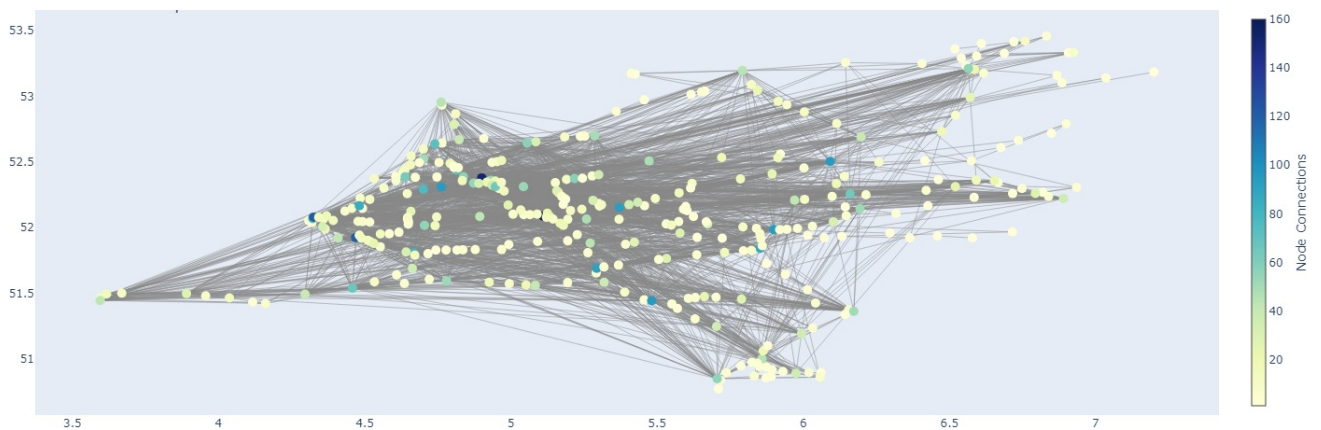
# Add node text
node_adjacencies = []
node_text = []
for node, adjacencies in G.adjacency():
    node_adjacencies.append(len(adjacencies))
    node_text.append(f'{node}<br># of connections: {len(adjacencies)}')

node_trace.marker.color = node_adjacencies
node_trace.text = node_text

# Create the figure
fig = go.Figure(data=[edge_trace, node_trace],
                layout=go.Layout(
                    title='Network Graph',
                    showlegend=False,
                    hovermode='closest',
                    margin=dict(b=0, l=0, r=0, t=0),
                    xaxis=dict(showgrid=False, zeroline=False),
                    yaxis=dict(showgrid=False, zeroline=False)
                ))

fig.show()

```



Creating a Daily Graph with Network Features

```

In [51]: # Get unique dates and stations
unique_dates = disruptions_df['Date'].dt.date.unique()
sources = disruptions_df['source'].unique()
destinations = disruptions_df['target'].unique()
stations = list(set(sources).union(set(destinations)))

# Function to create daily graph and calculate node features
def create_daily_graph(date, df):
    # Filter data for the specific date
    daily_df = df[df['Date'].dt.date == date]

    # Create graph
    G = nx.Graph()
    G.add_nodes_from(stations)

    # Add weighted edges
    for _, row in daily_df.iterrows():
        G.add_edge(
            row['source'],
            row['target'],
            weight=row['Rides planned'],
            distance=row['distance']
        )

    # Calculate node features
    node_features = {}
    for node in G.nodes():
        # Degree (number of connections)
        degree = G.degree(node)

        # Weighted degree (sum of rides planned)

```

```

        weighted_degree = sum(
            G[node][neighbor]['weight']
            for neighbor in G.neighbors(node)
        )

        # Average distance to neighbors
        avg_distance = (
            sum(G[node][neighbor]['distance'] for neighbor in G.neighbors(node)) / degree
            if degree > 0 else 0
        )

        node_features[node] = {
            'degree': degree,
            'weighted_degree': weighted_degree,
            'avg_distance': avg_distance
        }

    return G, node_features

# Dictionary to store daily graphs and features
daily_graphs = {}
daily_features = {}

# Process each day
for date in unique_dates:
    G, features = create_daily_graph(date, disruptions_df)
    daily_graphs[date] = G
    daily_features[date] = features

```

Example Graph for the First Day of the Dataset

```

In [ ]: # Example: Visualize graph for a specific date
def visualize_graph(date, G): # Position nodes

    edge_x = []
    edge_y = []
    for edge in G.edges():
        x0, y0 = pos[edge[0]]
        x1, y1 = pos[edge[1]]
        edge_x.extend([x0, x1, None])
        edge_y.extend([y0, y1, None])

    edge_trace = go.Scatter(
        x=edge_x, y=edge_y,
        line=dict(width=0.5, color='#888'),
        hoverinfo='none',
        mode='lines'
    )

    node_x = []
    node_y = []
    for node in G.nodes():
        if node not in pos:
            node_x.append(5.5)
            node_y.append(52)
            continue
        x, y = pos[node]
        node_x.append(x)
        node_y.append(y)

    node_trace = go.Scatter(
        x=node_x, y=node_y,
        mode='markers',
        hoverinfo='text',
        marker=dict(
            showscale=True,
            colorscale='YlGnBu',
            size=10,
            colorbar=dict(
                thickness=15,
                title=dict(
                    text='Node Connections',
                    side='right'
                ),
            ),
            xanchor='left'
        )
    )

    # Add node information
    node_text = []

```



```

node_colors = []
features = daily_features[date]
for node in G.nodes():
    info = (f"Station: {node}<br>"
            f"Degree: {features[node]['degree']}<br>"
            f"Rides: {features[node]['weighted_degree']}<br>"
            f"Avg Distance: {features[node]['avg_distance']:.2f}")
    node_text.append(info)
    node_colors.append(features[node]['weighted_degree'])

node_trace.text = node_text
node_trace.marker.color = node_colors

fig = go.Figure(
    data=[edge_trace, node_trace],
    layout=go.Layout(
        title=f'Train Network - {date}',
        showlegend=False,
        hovermode='closest',
        margin=dict(b=20,l=5,r=5,t=40),
    )
)

fig.show()

# Visualize for a specific date (example)
example_date = unique_dates[0]
visualize_graph(example_date, daily_graphs[example_date])

## Print features for first date as example
# print(f"Node features for {example_date}:")
# for node, features in daily_features[example_date].items():
#     print(f"{node}: {features}")

```

Node features for 2019-01-01:

```

Utrecht Terwijde: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Ommen: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Rhenen: {'degree': 3, 'weighted_degree': 34, 'avg_distance': 60.0}
Lage Zwaluwe: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
SSN Stoomdepot: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Bielefeld Hbf: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Roosendaal: {'degree': 8, 'weighted_degree': 55, 'avg_distance': 104.5}
Santpoort Zuid: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Diemen Zuid: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Driehuis: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Amersfoort Centraal: {'degree': 6, 'weighted_degree': 35, 'avg_distance': 91.5}
Lansingerland-Zoetermeer: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Rotterdam Stadion: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Rotterdam Beurs: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Baflo: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Brussel-Centraal: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Essen (B): {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Tiel Passewaaij: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Bourg-St-Maurice: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Hoorn Kersenboogerd: {'degree': 3, 'weighted_degree': 33, 'avg_distance': 135.0}
München Hbf: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Rotterdam Centraal: {'degree': 13, 'weighted_degree': 136, 'avg_distance': 103.6923076923077}
Hoorn: {'degree': 4, 'weighted_degree': 31, 'avg_distance': 96.5}
Bad Nieuweschan: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Amsterdam Muiderpoort: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Vlissingen Souburg: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Sittard: {'degree': 1, 'weighted_degree': 3, 'avg_distance': 19.0}
Offenburg: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Gilze-Rijen: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Wierden: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Capelle Schollevaar: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Zoetermeer: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Castricum: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Breda-Prinsenbeek: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Rotterdam Zuid: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Bussum Zuid: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Uitgeest: {'degree': 2, 'weighted_degree': 35, 'avg_distance': 54.5}
Delft Campus: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Spaubeek: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Dresden Hbf: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Freiburg (Breisgau) Hbf: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Hamm (Westf.): {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Helmond Brandevoort: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Voorhout: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Veenendaal West: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Hilversum: {'degree': 2, 'weighted_degree': 6, 'avg_distance': 23.5}
Eindhoven Strijp-S: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Mainz Hbf: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Hoevelaken: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}

```

Utrecht Leidsche Rijn: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Den Haag Moerwijk: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Waddinxveen Noord: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Sassenheim: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Wormerveer: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Barneveld Noord: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Cuijk: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Oldenzaal: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
IJlst: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
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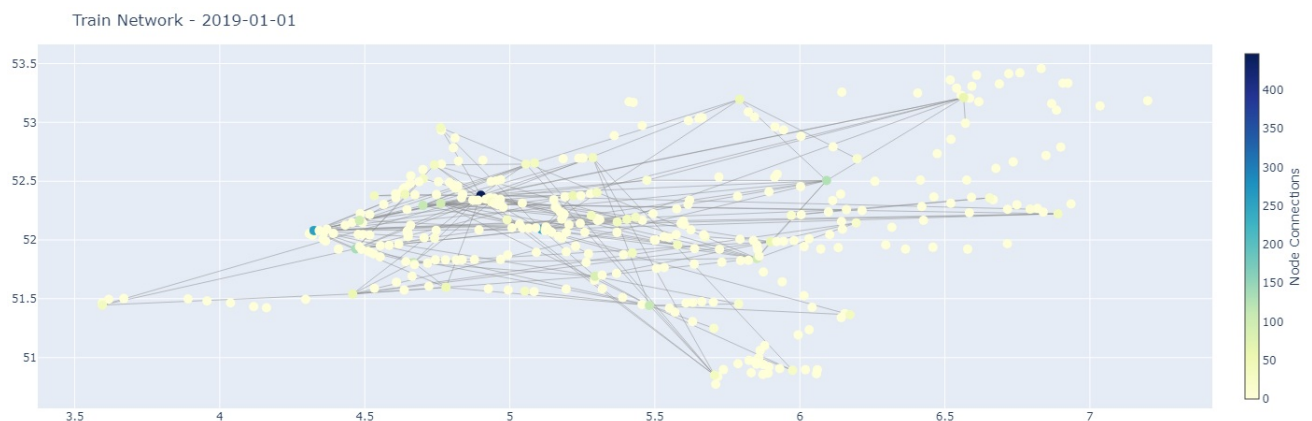
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Workum: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Almelo: {'degree': 2, 'weighted_degree': 17, 'avg_distance': 46.5}
Meppel: {'degree': 1, 'weighted_degree': 15, 'avg_distance': 67.0}
Brussel-Noord: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Arkel: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Geldrop: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Lochem: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
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Velp: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Best: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
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Diemen: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Sliedrecht Baanhoek: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
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London St. Pancras Int.: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Enschede: {'degree': 4, 'weighted_degree': 33, 'avg_distance': 158.25}
Sauwerd: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Deventer: {'degree': 4, 'weighted_degree': 4, 'avg_distance': 76.5}
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Nunspeet: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}

```

Arnhem Presikhaaf: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Tegelen: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Duisburg Hbf: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Den Haag Ypenburg: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
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Schin op Geul: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
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Marseille-St-Charles: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Bovenkarspel-Grootebroek: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
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Baarn: {'degree': 1, 'weighted_degree': 33, 'avg_distance': 24.0}
Houten Castellum: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Driebergen-Zeist: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Vorden: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Zürich HB: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Beverwijk: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Nijkerk: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Koblenz Hbf: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Wezep: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Brussel-Zuid/Midi: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
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Paris-Nord: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Varsseveld: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Steenwijk: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Echt: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Arnhem Zuid: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Leer (Ostfriesl): {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Stavoren: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Frankfurt Flughafen Fernb: {'degree': 0, 'weighted_degree': 0, 'avg_distance': 0}
Leeuwarden: {'degree': 5, 'weighted_degree': 49, 'avg_distance': 164.8}

```



Simple Logistic Regression for the First Month of the Dataset

Due to the imbalanced nature of the dataset, the model only seems to predict that a trajectory is not disrupted. This is a problem that will need to be resolved in the final model.

```

In [50]: # Assuming disruptions_df and daily_features are available from previous code
# Filter first month (assuming dates are in 2025)
start_date = pd.to_datetime('2019-01-01')
end_date = pd.to_datetime('2019-01-31')
first_month_df = disruptions_df[
    (disruptions_df['Date'] >= start_date) &
    (disruptions_df['Date'] <= end_date)
]
first_month_dates = first_month_df['Date'].dt.date.unique()

# Prepare feature dataset
feature_data = []
target = []

for date in first_month_dates:
    daily_df = first_month_df[first_month_df['Date'].dt.date == date]
    features = daily_features[date]

# For each source-target pair in the daily data

```

```

for _, row in daily_df.iterrows():
    source = row['source']
    destination = row['target']
    disrupted = row['Disrupted'] # Assuming this column exists

    # Combine features from source and target nodes
    combined_features = [
        features[source]['degree'],
        features[source]['weighted_degree'],
        features[source]['avg_distance'],
        features[destination]['degree'],
        features[destination]['weighted_degree'],
        features[destination]['avg_distance'],
        row['distance'], # Add edge distance as a feature
        row['Rides planned'] # Add rides planned as a feature
    ]

    feature_data.append(combined_features)
    target.append(disrupted)

# Convert to numpy arrays
X = np.array(feature_data)
y = np.array(target)

# Split into training and test sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Scale the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Train logistic regression classifier
classifier = LogisticRegression(random_state=42)
classifier.fit(X_train_scaled, y_train)

# Evaluate the model
y_pred = classifier.predict(X_test_scaled)
print("Classification Report:")
print(classification_report(y_test, y_pred))
print("\nConfusion Matrix:")
# Plot the confusion matrix
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['No Disruption', 'Disruption'], yticklabels=['No', 'Disruption'])
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()

# Feature importance (coefficients)
feature_names = [
    'source_degree', 'source_weighted_degree', 'source_avg_distance',
    'target_degree', 'target_weighted_degree', 'target_avg_distance',
    'distance', 'rides_planned'
]
print("Feature Coefficients:")
for name, coef in zip(feature_names, classifier.coef_[0]):
    print(f"{name}: {coef:.4f}")

```

Classification Report:

	precision	recall	f1-score	support
False	0.99	1.00	0.99	1724
True	0.00	0.00	0.00	24
accuracy			0.99	1748
macro avg	0.49	0.50	0.50	1748
weighted avg	0.97	0.99	0.98	1748

Confusion Matrix:

c:\Users\brake\UVA_AML24\week_1\.conda\Lib\site-packages\sklearn\metrics_classification.py:1565: UndefinedMetricWarning:

Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero\_division` parameter to control this behavior.

c:\Users\brake\UVA_AML24\week_1\.conda\Lib\site-packages\sklearn\metrics_classification.py:1565: UndefinedMetricWarning:

Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero\_division` parameter to control this behavior.

c:\Users\brake\UVA_AML24\week_1\.conda\Lib\site-packages\sklearn\metrics_classification.py:1565: UndefinedMetricWarning:

Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero\_division` parameter to control this behavior.



Feature Coefficients:

source_degree: -0.5219

source_weighted_degree: 0.7859

source_avg_distance: 0.7625

target_degree: 1.5596

target_weighted_degree: -1.2392

target_avg_distance: 0.4873

distance: -2.0096

rides_planned: -0.0377